

1st YEAR BCA SYLLABUS 1st Semester

Computer Fundamental & MS-Office (BCA-101)

Unit-I: Introduction to Computers: Introduction, Characteristics of Computers, Block diagram of computer. Types of computers and features, Mini Computers, Micro Computers, Mainframe Computers, Super Computers. Types of Programming Languages (Machine Languages, Assembly Languages, High Level Languages). Data Organization, Drives, Files, Directories. Types of Memory (Primary And Secondary) RAM, ROM, PROM, EPROM. Secondary Storage Devices (FD, CD, HD, Pen drive); I/O Devices (Scanners, Plotters, LCD, Plasma Display); Number Systems; Introduction to Binary, Octal, Hexadecimal system; Conversion, Simple Addition, Subtraction, Multiplication.

Unit-II: Algorithm and Flowcharts Algorithm: Definition, Characteristics, Advantages and disadvantages, Examples Flowchart: Definition, Define symbols of flowchart, Advantages and disadvantages, Examples.

Unit-III: Operating System and Services in O.S.: Dos – History, Files and Directories, Internal and External Commands, Batch Files, Types of O.S.

Unit-IV: Windows Operating Environment: Features of MS – Windows, Control Panel, Taskbar, Desktop, Windows Application, Icons, Windows Accessories, Notepad, Paintbrush.

Unit-V: Editors and Word Processors: Basic Concepts, Examples: MS-Word, Introduction to desktop publishing. Spreadsheets and Database packages: Purpose, usage, command, MS-Excel, Creation of files in MS-Access, Switching between application, MS-PowerPoint.

Introduction to Programming Using C (BCA – 102)

Unit-I: C basics: C character set, Identifiers and keywords, Data types, constants, variables and arrays, declarations, expressions statements, symbolic constants, compound statements, arithmetic operators, unary operators, relational and logical operators, assignment operators, conditional operators, bit operators.

Unit-II: Decision Control Structures: If statement, If-else statement, Nested if(), If() ladder, Switch, case statement, Iterative statements: for loop, while loop, Do-while () loop, Conditional statements: Break, Continue, Storage Classes, Array: Declaration of an Array, Initialization of Array, Types of Array: Single Dimension Array, Two, Dimensional Array, Address Calculation of an Element of a 2-D Array.

Unit-III: Functions: Library Functions, User Defined Functions, Function Declaration, Prototype Declaration, Types of Arguments: Actual Arguments, Formal Arguments, Function Definition. Passing Arrays as Parameters, Methods to Call a Function: Call by Value, Call by Reference.

Unit-IV: Pointers: Declaration of Pointer Variables, Pointer Arithmetic, Returning Multiple Output Values through a Function Strings.

Unit-V: Structures, Unions, Array of Structures, Enumerations, File Handling: Opening a file, Closing a file, File Opening Modes. Reading from and writing to a file, Copying Content of an existing file to another, Command line arguments, argc and argv Parameters, Pre-processor directives.

Business Communication and Soft Skills (BCA – 103)

Unit-I: Means of Communication - Meaning and Definition – Process – Functions – Objectives – Importance – Essentials of good communication – Communication barriers, 7C's of Communication, Types of Communication: Meaning, Nature & Scope.

Unit-II: Oral Communication: Principle of effective oral communication – Techniques of effective speech – Media of oral communication (Face-to-face conversation – Teleconferences – Press Conference – Demonstration – Radio Recording – Dictaphone – Meetings – Rumour – Demonstration and Dramatisation – Public address system – Grapevine – Group Discussion – Oral report – Closed circuit TV). The art of listening – Principles of good listening. **Written Communication - Purpose of writing, Clarity in Writing, Principle of Effective writing, Writing Techniques, Electronic Writing Process. Business Letters & Reports - Need and functions of business letters – Planning & layout of business letter – Kinds of business letters – Essentials of effective correspondence, Purpose, Kind and Objective of Reports, Writing Reports.**

Unit-III: Drafting of business letters - Enquiries and replies – Placing and fulfilling orders – Complaints and follow-up Sales letters – Circular letters Application for employment and resume. Information Technology for Communication - Word Processor – Telex – Facsimile(Fax) – E-mail – Voice mail – Internet – Multimedia – Teleconferencing – Mobile Phone Conversation – Video Conferencing – SMS – Telephone Answering Machine – Advantages and limitations of these types. Self Analysis: SWOT Analysis, Who am I, Attributes, importance of self confidence, self esteem, creativity: out of box thinking, Lateral thinking.

Unit-IV: Attitude: Factors influencing attitude, challenges and lessons from attitude, Etiquette, Motivation: Factors of motivation, Self talk, Intrinsic & Extrinsic Motivators. Goal Setting: Wish List, SMART Goals, Blue print for success, Short Term, Long Term, Life Time Goals. **Interpersonal Skills:** Gratitude: Understanding the relationship between leadership networking & team work. Assessing Interpersonal skills situation description of interpersonal skills, Team work: Necessity of team work personally, socially, and educationally.

Unit-V: Leadership: Skills for a good leader, Assessment of leadership skills, Stress Management: Causes of stress and its impact, how to manage & distress, Circle of control, Stress Busters, Emotional Intelligence: What is Emotional Intelligence, emotional quotient why emotional intelligence matters, Emotion Scales, Managing Emotions. **Conflict Resolution:** Conflict in human relations – Reasons case studies, Approaches to conflict resolution. **Decision Making:** Importance and necessity of Decision Making, Process and practical way of Decision Making, Weighing Positives and Negatives.

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Introduction to HTML, CSS - XML (BCA – 104)

Unit-I: Basics of Internet and Web: The basics of Internet, World Wide Web, Web page Home page, Web site, Static, Dynamic and Active web page, Overview of Protocols, Simple Mail Transfer Protocol, Gopher, Telnet, Emails, TFTP, Simple Network Management Protocol, Hyper Text Transfer Protocol, Client server computing concepts. Web Client and Web Server, Web Browser, Browser e.g., Netscape navigator, Internet Explorer, Mozilla Firefox, Client-Side Scripting Languages, VB Script and Java Script, Active X control and Plug-ins, Web server Architecture, Image maps, CGI, API web database connectivity, DBC, ODBC.

Unit-II: Dynamic HTML, Document Object Model, Features, DHTML, CSSP (Cascading Style Sheet Positioning) and JSSS (Java Script assisted Style Sheet), Layers of Netscape, The ID Attribute, DHTML Events.

Unit-III: Introduction to HTML: Editors, Basics, Elements, Attribute, Headings, Paragraphs, Styles, Formatting, Quotations, Comments, CSS, Links, Images, Tables, Lists, Blocks, Classes, ID, frames, File Paths, Head, Layout, Computer Code, Entities, Symbols, Char set, Color and Background of Web Pages, Hypertext, Hyperlink and Hypermedia, Links, Anchors and URLs, Links to External Documents, Different Section of a page and graphics, Footnote and E-mailing, Creating Table, Frame, Form and Style Sheet.

Unit-IV: CSS: Introduction, Syntax, Colors, Backgrounds, Borders, Margins, Padding, Height/Width, Box Model, Outline, Text, Fonts, Icons, Links, Lists, Tables, Display, max, Width, Position, Overflow, Float, Inline, Block, Align, Combinators, Pseudo, Class, Pseudo Elements, Opacity, Navigation Bar, Dropdowns, Image Gallery, Image Sprites, Attr Selectors, Forms, Counters, Website Layout, Units, Specificity.

Unit-V: XML: Introduction, Tree, Syntax, Elements, Attributes, Namespaces, Display, HTTP request, Parser, DOM, XPath, XSLT, XQuery, XLink, Validator, DTD, Schema, Server.

Mathematics-I (BCA – 105)

Unit-I: Determinants: Definition, Minors, Cofactors, Properties of Determinants. **MATRICES:** Definition, Types of Matrices, Addition, Subtraction, Scalar Multiplication and Multiplication of Matrices, Adjoint, Inverse, Cramers Rule, Rank of Matrix Dependence of Vectors, Eigen Vectors of a Matrix, Caley-Hamilton Theorem (without proof).

Unit-II: Limits & Continuity: Limit at a Point, Properties of Limit, Computation of Limits of Various Types of Functions, Continuity at a Point, Continuity Over an Interval, Intermediate Value Theorem, Type of Discontinuities.

Unit-III: Differentiation: Derivative, Derivatives of Sum, Differences, Product & Quotients, Chain Rule, Derivatives of Composite Functions, Logarithmic Differentiation, Rolle's Theorem, Mean Value Theorem, Expansion of Functions (Maclaurin's & Taylor's), Indeterminate Forms, L'Hospital's Rule, Maxima & Minima, Curve Tracing, Successive Differentiation & Leibnitz Theorem.

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Unit-IV: Integration: Integral as Limit of Sum, Fundamental Theorem of Calculus (without proof.), Indefinite Integrals, Methods of Integration Substitution, By Parts, Partial Fractions, Reduction Formulae for Trigonometric Functions, Gamma and Beta Functions (definition).

Unit-V: Vector Algebra: Definition of a vector in 2 and 3 Dimensions; Double and Triple Scalar and Vector Product and physical interpretation of area and volume.

IInd Semester

Object Oriented Programming Using C++ (BCA – 201)

Unit-I: Introduction: Introducing Object Oriented Approach, Procedural Programming Language Vs. Object Oriented Language, Basic concept of OOPs, Operators, Tokens, variables, keywords, Data types, identifiers, characters, type def statement, constants, Enumerated data type.

Unit-II: Control Flow: If statement, If else statement, Nested If, Else, Statements, For Loop, While Loop, Do, While Loop, Break, Switch, Continue, goto. Classes and Objects, Encapsulation, information hiding, abstract data types, object & classes, attributes, methods, C++ class declaration, Constructors and destructors, Default parameter value, object types, C++ garbage collection, dynamic memory allocation, Metaclass / abstract classes.

Unit-III: Array: Array Illustration, Multi, Dimensional arrays, Strings, Array of strings, function prototype, function return data type, parameter passing, default argument, Inline function, Function Overloading, Array Function, Operator Overloading.

Unit-IV: Pointers: Pointer to Derived Class, array of pointers, Inheritance and Polymorphism: Inheritance, Class hierarchy, derivation, public, private & protected, abstract classes, Single, Multilevel, Multiple, Hierarchical, Hybrid, benefits of Inheritance.

Unit-V: Files and Exception Handling: Streams and files, Namespaces, Exception handling.

Digital Electronics (BCA – 202)

Unit-I: Number System & Boolean Algebra: Number system: Binary, Octal, Decimal, Hexadecimal, Conversion of Number System, Binary Arithmetic & Complement, Binary Codes: Weighted & Non Weighted, Gray Code, Excess-3 Code. Error Detection Codes, Hamming Code, Boolean Function, Boolean Postulates, De-Morgan's Theorem, Boolean Expressions: Sum of Product, Product of Sum, Minimization of Boolean Expressions, using K-Map, Logic Gates: AND, OR, NOT, NAND, NOR, XOR, XNOR, Implementations of Logic Functions using Gates, NAND, NOR, Implementations, Multilevel gate Implementations.

Unit – 2: Combinational Circuits: Adders & Subtractors: Half adder, full adder, binary adder, half subtractor, full subtractor, magnitude comparator: Two Bit comparator, three Bit Magnitude comparator, multiplexer & De-multiplexer: 4*1 Multiplexer, 8*1 Multiplexer, Decoder & Encoder, parity Checker & Generator, code converter.

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Unit – 3: Sequential Circuit: Introduction to flip flops: SR, JK, T, D master Slave flip, conversion of flip flops, characteristics table & Equation, edge triggering & Level Triggering, excitation table.

Unit – 4: Registers: Introduction of Registers, classification of Registers, register with parallel load, shift registers, bidirectional shift register with parallel load.

Unit – 5: Counters: Introduction of counter, Asynchronous/Ripple counter, synchronous counters, BCD counter, 4 – bit binary counter with parallel load, design of synchronous counters, ring counter, Johnson counter.

Data Structure Using 'C'/C++ (BCA – 203)

Unit – 1: Classification of Data structure: Operations on Data Structure, Address Calculating, Application of Array, Limitation of Array, application of Arrays, Array as parameters, sparse Matrices

Unit – 2: Continuous Implementation (Stack): Array Representation, Operations of Stacks; Push & Pop, Applications of stack, Conversion of Infix to Prefix and postfix Expressions, Evaluation of postfix expression using stack
Recursion: Recursive Definition and process, principles of Recursion, Tower of Hanoi problem, recursion Vs. Iteration
Continuous. Implementation (Queue): Array representation and implementation of Queues, operations on queue: Create, Add, Delete, full and empty queue, circular queue, De-queue and priority queue.

Unit – 3: Non Continuous Implementation Link Lists: Linear list concept, linked list terminology representation of Linked list in memory, types of linked list, single linked list, doubly linked list, single circular list, circular doubly linked list, operations on link list: Create list insert node (empty list, beginning, middle, end), delete node (first, general case). Traversing node, searching node, print list, count nodes, sort lists.

Unit – 4: Trees: Introduction to tree & its Terminology, binary trees types of Binary trees representation of binary tree, traversals (inorder, preorder, postorder), tree expression, binary search tree, insertion and deletion in BST

Unit – 5: Sorting & searching Techniques: Bubble sort, selection sort, insertion sort, quick sort, merge sort, sequential search, and binary search.

Principal of Management (BCA – 204)

Unit-I: Nature of Management - Meaning, Definition, it's nature purpose, importance & Functions, Management as Art, Science & Profession. Management as social System Concepts of management-Administration-Organization, Evolution of Management.

Unit-II: Functions of Management: Planning – Meaning- Need & Importance, types levels, advantages & limitations. Forecasting- Need & Techniques; Decision making-Types - Process of rational decision making & techniques of decision making Organizing.

Unit-III: Elements of Organizing & Processes: Types of organizations Delegation of authority – Need, difficulties; Delegation – Decentralization Staffing – Meaning & Importance; Direction – Nature – Principle. Communication – Types & Importance. Motivation – Importance – theories Leadership – Meaning – styles, qualities & function of leader.

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Unit-IV: Functions of Management: Controlling - Need, Nature, importance, Process & Techniques. Coordination – Need – Importance. Strategic Management - Definition, Classes of Decisions, Levels of Decision, Strategy, Role of different Strategist, Relevance of Strategic Management and its Benefits, Strategic Management in India.

Unit – V: Recent Trends in Management: Social responsibility of management - environment friendly management, Management of Change, Management of Crisis, Total Quality Management, Stress Management, International Management.

Numerical Methods (BCA - 205)

Unit-I: Roots of Equations: Bisections Method, False Position Method, Newton's Raphson Method, Rate of convergence of Newton's method.

Unit-II: Interpolation and Extrapolation : Finite Differences, The operator E, Newton's Forward and Backward Differences, Newton's dividend differences formulae, Lagrange's Interpolation formula for unequal Intervals, Gauss's Interpolation formula, Starling formula, Bessel's formula, Laplace-Everett formula.

Unit-III: Numerical Differentiation Numerical Integration : Introduction, direct methods, maxima and minima of a tabulated function, General Quadratic formula, Trapezoidal rule, Simpson's One third rule, Simpson's three- eight rule.

Unit-IV: Solution of Linear Equation: Gauss's Elimination method and Gauss's Siedel iterative method.

Unit-V: Solution of Differential Equations: Euler's method, Picard's method, Fourth-order Runga – Kutta method.

IInd YEAR BCA SYLLABUS IIIrd Semester

Data Base Management System (C -301)

Unit – I: Introduction: Database system concepts, Database Users, and Architecture Introduction to Database System with example, Introduction to Traditional File Oriented System, Characteristics of the Database Approach, Components of Database System, Database users, Advantages and disadvantages of using a DBMS, structure of DBMS, Database Schemas and Instances, DBMS Architecture, Data Independence, Database Languages and Interfaces, Classification of Database Management Systems.

Unit – II Data Modeling & Relational Database Management System Data Modeling using the Entity Relationship Model: Entity Types, Entity Sets, Attributes, Keys, Relationships, Relationship Types, Role and Structural, Constraints, weak Entity Types, ER Diagrams, Naming Conventions, Design issues.

Unit – III The Relational Data model: Relational Constraints and the Relational Algebra: Relational Model Concepts, Relational Constraints and Relational database Schemas update Operations and Dealing with Constraint Violations, Basic Relational Algebra Operations, Additional Relational Operations, and Examples of Queries in Relational Algebra.

Unit – IV :SQL: SQL and Database Design Theory and Methodology Structured Query Language The Relational Database Standard: Data Definition, Constraints and schema Changes in SQL, Types of SQL Commands, SQL Operators and their Procedure, insert, Delete, and Update Statements in SQL Queries and Sub Queries, Aggregate Functions, Joins, Unions, Intersection, Minus Views (Virtual Tables) in SQL. Functional Dependencies and Normalization of Relational Database: Informal Design Guidelines for Relation Schemas, Functional Dependencies, Armstrong Rules, Closure of Attributes, Normal Forms Based on Primary Keys, General Definitions of Second and Third Normal Forms, Boyce Codd Normal Form.

Unit– V Transaction Processing: Concurrency Control and Distributed Database Transaction Processing Concepts: Introduction to Transaction Processing, Transaction and system concepts, Desirable Properties of Transactions, Concurrency Control Techniques, Locking Techniques for concurrency Control, Concurrency Control Based on Timestamp Ordering.

E – Commerce and ERP (C -302)

Unit – I : Introduction: Defining E – Commerce, Main Activities of Electronic Commerce, Benefits of E – Commerce, Goals of Electronic Commerce, Main components of E – Commerce, Functions of Electronic Commerce, Communication, Process Management, Service Management, Transaction Capabilities, Process of E – Commerce, Types of E – Commerce, Role of internet and web in E – Commerce, Technologies Used in E – Commerce Systems, Scope of E – Commerce, E – Business Models.

Unit – II : E – Commerce Activities: Various Activities of E – Commerce, Various Modes of operation Associated with E – Commerce, Matrix of E – Commerce Types, Elements and Resources Impacting E – Commerce and Changes, Type of E – Commerce Providers and Vendors, Man Power Associated with E – Commerce Activities, Opportunity Development for E – Commerce Stages, Development of E – Commerce Business Case, Components and Factors for the Development of the Business Case, Steps to Design and Design and Develop and E – Commerce Website.

Unit – III Internet: The Backbone for E – Commerce: Early Ages of internet, Networking Categories, Characteristics of Internet, Components of Internet, Internet Services, Elements of Internet, uniform Resource Locators, Internet protocol, Shopping Cart, Cookies and E – Commerce, Web Site Communication, Strategic Capabilities of internet. Implementation of E – Commerce: WWW, EBAY, COM, B2B website – registration, Time factor, Bidding process, Growth of eBay, Paypal, New Trend in making Payments Online- National Electronic Fund Transfer.

Unit – IV ISP, WWW and Portals: Internet Service Provider (ISP), World Wide Web (WWW), Portals, Steps to build homepage, Metadata, Advantages of Portal, Enterprise Information Portal (EIP), E – Marketing: Traditional Marketing, E – Marketing. Identifying Web Presence Goals, Achieving Web Presence goals, Uniqueness of the Web, Meeting the needs of website visitors, Maintaining a website, Metrics Defining internet units of Measurement, Online Marketing, Advantages of online Marketing, Content:

Format and access, Maintaining a website. Metrics Defining Internet Units of Measurement Online Marketing, Advantages of Online Marketing, E – Security: Security on the internet, Network and website Security Risks, Denial of service attacks, viruses, unauthorized access to a computer network, vulnerability of internet sites, network and website security, Transaction security and data protection, Security audits and penetration testing, E- Business Risk Management issues, Firewall, Network, Policy, Advanced authentication mechanism, Packet filtering, Application gateways, Defining enterprise Wide Security Framework.

Unit – V E – Payment Systems: Electronic Funds Transfer, Digital Token Based E – Payment Systems, Modern Payment Systems, Steps for Electronic Payment, Payment Security, Net Banking, Customer Relationship Management: Customer Relationship Management (CRM), Marketing automation, Enterprise customer management, customer relationship Management Areas, CRM Processes, Event triggers, business logic and rules repository, Decision support tools, Higher level statistical analysis, Collateral management, Electronic Customer Relationship Management, Need, Architecture and Applications of electronic CRM.

Computer Organization and Architecture (C -303)

Unit –I Computer Evolution: Brief history of Computer, Classification of computer, Structure of a computer system, Arithmetic Logic Unit, Control Unit, Von Neumann Architecture, Integer Addition and Subtraction, Floating point representation, Signed numbers, Binary Arithmetic 1's and 2's Complements, Booths Algorithm, Hardware Implementation, IEEE Standards, Floating Point Arithmetic, The accumulator, shifts, carry and overflow instruction Characteristics, CPU with single BUS, Types of operands, Types of operations, Addressing Modes, instruction Formats.

Unit –II Processor Organization: Parallelism and Computer arithmetic, Computer arithmetic associatively, Floating Point in the 8086, Programmers Model of 8086, Register Organization, 8086 Registers, Instruction Cycles, Addressing Modes, Micro operations, The instruction cycle, Control of the CPU, Functional Requirements, Single, Two, three bus structure, Execution of a complete instruction, Branching, Sequencing of Control Signals, Hardwired Control Unit, Micro – Programmed Control.

Unit –III Memory Organization: Characteristics of Memory Systems, Main Memory, Types of memory, Memory system considerations, Design of memory subsystem using Static, Dynamic Memory Chips, Memory interleaving High Speed Memories: Cache Memory, Structure of cache and main memory, Elements of Cache Design, Mapping functions, Replacement algorithms, External Memory, Virtual memory.

Unit –IV I/O Organization: Input/ Output Module: Need, Techniques, Interrupt Driven I/O, Basic concepts of an Interrupt, Response of CPU to an Interrupt, Design issues, Priorities, interrupt handling, Types of interrupts. Data transfer Techniques, Data memory Access, Buses, Types of buses, I/O interface, synchronous and Asynchronous Data Transfer, serial I/O, Input Devices, Output Devices, Multi-programming vs. Multiprocessing, Comparison between closely coupled and closely coupled multiprocessor.

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Unit – V Micro-programming: Basic Principles, Features, Hardwired vs. Micro programmed computers, Applications and advantages of micro programming, Limitations of microprogramming, Computer Clock, Micro instruction s and its Control Path, Microcode, Machine instruction. Parallel Organization, Instruction Set Architecture (ISA), RISC and CISC, Characteristics of CISC, Characteristics of RISC, RISC versus CISC, vector Processing requirements and characteristics of vector processing.

Operating System with the case Study of UNIX and Windows (C -304)

Unit – I Introduction: What is an operating system, Simple Batch Systems, Multi, Programmed Batch System, Time, Sharing System, Personal, Computer systems, Parallel Systems, Distributed Systems, Real, Time Systems, Memory Organization: Fixed memory ,, variable memory, Memory Management: Background, Logical versus physical Address space, swapping, Contiguous allocation, paging, Segmentation Virtual memory: Demand Paging, Page Replacement, Page replacement Algorithms, Performance of Demand Paging, Allocation of Frames, thrashing.

Unit – II Processes: Process Concept, Process Scheduling, operation on processes, CPU Scheduling: Basic Concepts, Scheduling Criteria, Scheduling Algorithms, Multiple, Processor Scheduling, Process Synchronization: Background, the critical, section problem, synchronization Hardware, Semaphores, Classical Problems of synchronization. Deadlocks: system Model, Deadlock Characterization, Methods for Handling Deadlocks, Deadlock Prevention, Deadlock Avoidance, Deadlock Detection, Recovery from Deadlock.

Unit – III Device Management: Techniques for Device Management, Dedicated Devices, Shared Devices, Virtual Devices, Input or Output Devices, Storage Devices, Buffering, Secondary Storage Structure: Disk Structure, Disk Scheduling, Disk Management, Swap, Space Management, Disk Reliability

Information Management: Introduction, A simple file system, General Model of a File system, Symbolic File System, Basic File System, Access Control Verification, Logical File System, Physical File System File System Interface, File Concept, Access Methods, Directory Structure, Protection, Consistency Semantics File, System Implementation: File, System Structure, Allocation Methods, Free Space Management.

Unit- IV Unix: A Sample Login Session, Logging On, Using the on – line Man pages, Logging Off, Directory and File Structure, File Names, Directories, The df Program, Your Login Directory, Subdirectories, Specifying Files, Protecting Files and Directoreis, Text Editors, Files as Output and Log files, Logging Your Actions to a File, Comparing Files, Searching Through Files, The Systems and Dealing with Multiple Users.

Unit – V Windows: Features of windows desktop, start menu, control panel, my computer, windows explorer, accessories, Managing Multiple Windows, arranging icons on the desktop, creating and managing folders, managing files and drivers, logging off and shutting down windows Entertainment CD player, VD Player, media player, Sound recorder, volume Control.

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Statistical Method and Application ((C -305))

Unit – I Classification of data, Tabulation of data, preparation of frequency distribution, presentation of data through histogram, frequency polygon curve.

Unit – II Measures of Central Tendency: Computer of Arithmetic Mean, Median and mode for ungrouped data grouped data, verification of median through ogives.

Unit – III Measures of Dispersion: Computation of Range, Quartile deviation, mean deviation and standard deviation, coefficient of variation. (Numerical Application only)

Unit – IV Concept of skewness, Karl Pearson's and Bowley's Coefficients of skewness (Numerical application only)

Unit – V Meaning of Correlation, Types of correlation, Correlation Coefficient, Karl Pearson, Spearman's rank correlation coefficient. (Numerical Application only).

IVth Semester

Java Programming (C - 401)

Unit – I Introduction, Java Tokens, Java Statements, Command Line Arguments, Programming Style Constant, Variables and Data Types Constants, Variables, Data Types, Declaration of variables, Giving Values of variables, scope of variables, symbolic constants, Type Casting, Getting values of variables, Standard Default values, Java Program Structure, Java Virtual Machine.

Unit – II Operators, Expressions and Statements: Arithmetic Operators, Relational Operators, Logical Operators, Assignment Operators, Increment and Decrement Operators, Conditional Operators, Bitwise operators, Special Operators, Arithmetic Expressions, Evolution of Expressions, Precedence of Arithmetic Operators Decision making and Branching: Introduction, Decision Making with if statement, Simple if Statement, The if... else Statement, Nesting of if else Statements, else if ladder, switch Statement?: Operator, Decision Making and Looping: Introduction, while statement, do statement, for statement.

Unit – III Classes, Objects and Methods: Defining a Class, Creating objects, Accessing Class Members, Constructors, Methods overloading, Static Members, Nesting of Methods, Inheritance Extending a Class, Overriding Methods, Final Variables and Methods, Final Classes, Finalize Methods Abstract Methods and classes, Visibility Control, Arrays One, Dimensional Arrays, Creating an Array, Two Dimensional Arrays, Strings, Vectors, Wrapper Classes.

Unit – IV Interfaces and Packages: Introduction, Defining interfaces, Extending interfaces, implementing interfaces, Accessing interface variance, Packages: Introduction, Java API packages, using system packages, Naming Conventions, Creating packages, Accessing a Packages, using a packages, Adding a Class to a Package, Hiding Class. Multithreaded Programming: Introduction, Creating Threads, Extending the Threads Class, Stopping and Blocking a Thread, Life Cycle of a Thread, Using Thread Methods, Thread Exception, Thread Priority and Synchronization.

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Unit – V Applet Programming: Introduction, How Applets Differ from Application, Preparing to write Applets, Building applet Code, Applet Life Cycle, Creating an Executable Applet, Designing a Web Page, Applet Tag, Adding Applet to HTML File, Running the Applet, More About Applet Tag, Managing Errors and exceptions: Statements, Using Finally Statement, Throwing Our Own Exceptions, Using Exceptions for Debugging.

Web Technology using PHP and MYSQL (C -402)

Unit – I PHP: Introduction to PHP Evaluation of PHP, Basic Syntax, Defining variable and constant, PHP data type, Operator and Expression, Decisions and loop making decisions, Doing repetitive task with looping, Mixing decisions and looping with KHTML, Function: Define a function, Call by value and call by reference, Recursive function, String Creating and accessing, String Searching & Replacing string, Formatting string, String related Library function.

Unit – II Array: Anatomy of an Array, Creating index based an Associative array Accessing array Element Looping with index based array, looping with associative array using each () and Foreach (), Some useful library function, Handling HTML Form PHP capturing Form Data dealing with multi –value filled, and generating file uploaded form, redirecting a form after submission.

Unit – III Working with file and Directories: Understanding file & directory, opening and closing, a file, coping, renaming and deleting a file, working with directories, creating and deleting folder, file uploading & Downloading.

Unit – IV Session and Cookie: Introduction to Session Control, Session Functionality what is a Cookie, Setting Cookies with PHP, using Cookies with Sessions, Deleting Cookies, Registering Session Variables, Destroying the variable and Session 8 Database Connectivity with

Unit – V MySQL : Introduction to RDBMS, Connection with MySQL Database, Performing basic database operation (DML) (insert, Delete, update, Select), Setting query Parameter, Executing query join (Cross joins, inner joins, outer joins, self joins) Exception Handling Understanding Exception and error, Try, Catch, Throw, Error tracking and debuting.

Artificial Intelligence (C -403)

Unit – I At Concepts, Various definitions of AI, Knowledge, knowledge pyramid, People and computers, what computers can do better than people, what people can do better than computers, characteristics of AI problems, Problem Representation in AI, Components of AI, AI Evolution, Application Areas of AI, History of AI, the Turing Test and the Revised Turing Test.

Unit – II Expert System: Components of Expert System: Knowledge Base, Inference Engine, User interface, Features of Expert System, Expert System Life Cycle, Categories of Expert System, Rule Base Vs. Model Based Expert Systems, Advantages/Limitations of Expert System, Developing an Expert System, Identification, Conceptualization, Formulation, Implementation, Testing, Using an Expert System, Application Areas of Expert System.

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Unit – III AI and Search Process: Brute Force Search, Depth First/ Breadth First Search, Heuristic Search: Hill Climbing, Constraints Satisfaction, Mean End Analysis, Best First Search, A* Algorithm, AO* Algorithm, Beam Search.

Unit – IV Natural Language Processing: Introduction, Need, Goal, Fundamental Problems in Natural Language Understanding, How People overcome Natural Language Problems, Speech Recognition: Introduction, Advantages and Approaches, Introduction to Robotics: Parts of a Robot Controlling.

Unit – V Applications: Communication, Communication as action, Formal Grammar for a fragment of English, Syntactic analysis, Augmented grammars, Semantic interpretation, Ambiguity and disambiguation, Discourse understanding, Grammar induction, Probabilistic language processing, probabilistic language models, information retrieval, information Extraction, Machine Translation.

Computer Network (C -404)

Unit – I Introduction: Definition of a Computer Network, Components of a computer network, Types of Network Based on Topology (Bus, Star, Ring Mesh Tree), Based on size Technology and ownership (LAN, MAN, WAN), Network topologies, Linear Bus Topology, Ring Topology, Star Topology, Hierarchical or Tree Topology, Topology comparison Considerations when choosing a Topology Switching, Circuit switching Message switching, Packet switching implementation of packet switching, Relationship between Packet Size and Transmission Time. Comparison of switching techniques: Multiplexing FDM, Frequency division multiplexing WDM, Wavelength division multiplexing, TDM Time division multiplexing.

Unit – II Network Software & Network Standardization: Introduction software protocol hierarchy, Design issues for the layers, merits and de – merits of layered Architecture, Service primitives: Reference models, the OSI reference Model, the TCP/IP reference Model, Comparison of the OSI & TCP/ IP Reference Model.

Unit – III Data Link Layer: Services provided to the upper layer, Framing Error Control, Flow control, IEEE Standards for MAX Sub layer, Network Layer, Services provided to the upper Layer Routing Algorithm (Centralized, Distributed), congestion Control (Token Based and Non Token Based) Internetworking.

Unit – IV Data Communications Introduction: Theoretical basis for communication, Fourier analysis, Band limited signals, maximum data rate of a channel: Transmission impairments, Attenuation distortion, Delay distortion, Dispersion, Noise: Data transmission modes, Serial & Parallel, simplex, Half duplex & full duplex, synchronous & Asynchronous transmission:

Unit – V Transmission medium: Introduction: Transmission medium, Guided & unguided Transmission medium, Twisted pair, coaxial cable, Optical fiber, Comparison of fiber optics and copper wire: Wireless transmission.

Optimization Techniques (C -405)

Unit – I Basic of operation research (OR): Characteristics of OR, Necessity of OR in industry, OR and decision making, role of computers in OR, Linear programming: Formulations and graphical solution of (2 variables) canonical and standard terms of linear programming problem.

Unit – II Algebraic solution: Simplex methods, Charnes method of penalties, two phase simplex method.

Unit – III Transportation model: Definition, Formulation and solution of transportation models, the row, minima column, minima matrix, minima and Vogel's approximation methods, Assignment model: Definition of assignment model, comparison with transportation model, formulation and solution of assignment model.

Unit – IV Sequencing Problem: Processing of n jobs through 2 machines, processing n jobs through 3 machines, processing 2 jobs through m machines.

Unit – V Game theory: Characteristics of games, maxima, minimax criteria of optimality, dominance property, algebraic and graphical method of solution of solving 2×2 games.

IIIrd YEAR BCA SYLLABUS

Vth Semester

Network Security (C -501)

Unit – I Network Security: Introduction: OSI security Architecture – Classical Encryption techniques Cipher principles, Data Encryption Standard, Block Cipher Design Principles and Modes of Operation.

Unit – II Public Key Cryptography: Key Management, Diffie Hellman Key Exchange – Elliptic Curve Architecture and Cryptography, introduction to number Theory, Confidentially using symmetric Encryption, public key cryptography and RSA.

Unit – III Authentication and Hash Function: Authentication requirements, Authentication functions – Message Authentication Codes, Hash Functions, Security of Hash Functions and MACs MDS message Digest algorithm, secure Hash algorithm, RIPEMD, HMAC Digital signatures, Authentication protocols, Digital Signature standard

Unit – IV Network Security: Authentication Applications: Kerberos, X.509 Authentication service, Electronic Mail Security, PGP, S/MIME, IP Security, Web Security.

Unit – V : System Level Security: Intrusion detection, password management, Viruses and related Threats, Virus Counter measures, Firewall Design Principles, Trusted systems.

Visual Basic .NET (C -502)

Unit – I Visual Basic .NET and the .NET Framework: Introduction to .net framework, Features, common Language Runtime (CLR), Framework class Library (FCL), Visual Studio.Net – IDE, Languages Supported, Components, Visual Programming, VB.net, Features, IDE, Menu system, Toolbars code Designer, Solution Explorer, Object Browser, Toolbox, Class View Window, properties Window, server Explorer, Task List, Output Window, Command Window.

Unit –II Elements of Visual Basic .net: Properties, Events and Methods of Form, Label, Text Box, List Box, Combo Box, Radio Button, Button, Check Box, Progress Bar, Date Time Picker, Calendar, Picture Box, H Scroll Bar V Scroll Bar, Group Box Tool Tip, Timer.

Unit –III Programming in Visual basic.net: Date types, Keywords, Declaring variables and constants, Operators, Understanding scope and accessibility of variable, Conditional Statements, if –then, If –then, Else,, Nested if, Select Case Looping Statement, Do loop, for Each, Next Loop, while Loop, Arrays, Static and Dynamic.

Unit – IV Functions, Built –in Dialog Boxes, Menus and toolbar: Menus and toolbars, Menu strip, Tool Strip, Status Strip, Built – in – Dialog Boxes – Open File Dialog, Save File Dialogs, Font Dialogs, Color Dialogs, Print Dialogs, Input Box, MsgBox, Interfacing with end use, Creating MDI parent and Child, Functions and procedure, Built –in Functions, Mathematical and String Functions, user Defined functions and procedures.

Unit – V Advanced Concepts in VB.Net: Object Oriented programming, Creating Classes, Object, Fields, Properties, Methods, Events, Constructors and destructors, Exception Handling, Models, Statements, File Handling, Using File Stream Class, File Model, File Share, File Access Enumerations, Opening or Creating Files with File Stream Class, Reading and writing Text using Stream Reader and Stream Writer Classes, Data Access With ADO.net – What are Databases? Data Access with server Explorer, Data Adapter and Data Sets, ADO.NET objects and Basic SQL.

Computer Graphics (C -503)

Unit – I Introduction: The Advantages of interactive Graphics, Representative uses of Computer Graphics Classification of Application Development of Hardware and software for computer Graphics conceptual Framework for Interactive Graphics: Overview, Scan Converting Lines, Scan Converting Circles, and Scan converting Ellipses.

Unit – II Hardcopy Technologies, Display Technologies, Raster, Scan Display System, Video Controller, Random Scan Display processor, input Devices for Operator interaction, image scanners, working exposure on graphics tools like Dream Weaver, 3D Effects etc, Clipping southland, Cohen Algorithm, Cyrus, Beek Algorithm, Midpoint subdivision algorithm.

Unit –III Geometrical Transformation: 2D Transformation, Homogeneous Coordinates and Matrix Representation of 2D Transformations, composition of 2D Transformations, the window to viewport Transformations, introduction to 3D Transformations Matrix.

Unit –IV Representing Curves & Surfaces: Polygon meshes parametric, Cubic Curves, Quadric surface, Solid Modeling, Representing Solids, Regularized Boolean Set Operation Primitive instancing Sweep Representations, Boundary Representations, Spatial Partitioning Representations, Constructive Solid Geometry comparison of Representations.

Unit – V Introductory Concepts: Multimedia Definition, CD – ROM and the multimedia highway, Computer Animation (Design, type of animation, using different functions) Unit – VI Uses of Multimedia, Introduction to making multimedia, The Stage of project, Hardware & Software requirements to make good multimedia skills and Training opportunities in Multimedia Motivation for Multimedia usage.

System Analysis and Design (C -504)

Unit – I Overview of Systems Concepts, Analysis and Design Life Cycle, Introduction to System concept: Characteristics of the system, Elements of a System, Types of Systems, Physical and Abstract System, Open and Closed System, Formal and informal system, introduction to Data and information: Types of information system, Categories of information system, Needs of Information Systems, Qualities of information system, Software Development Life Cycle (SDCL), Role and Attributes of system Analyst.

Unit – II System Planning and Requirements Determination System planning and initial investigation: Strategic plan for information processing, Tools for planning, Problems in planning, Need for requirement definition.

Unit – III Information gathering tools: Review of Literature, Procedures and forms, Methodologies, Tools and Techniques of Analysis Systems Analysis and Design Tree, Data Dictionary, Decision Table, Structured English, Data Flow Diagram, Components of a DFD Transformation and Decomposition, Context Diagram, Leveling a DFD, Feasibility study: Economic Feasibility (Cost & Benefit Analysis), Organizational Feasibility, Technical Feasibility, Behavioural Feasibility study

Unit – IV System Design and Implementation Process of Design: Logical and Physical Design, Design Methodologies, Elements of Form Design of Form Design, Design of Output, Design of File, Design of procedure, Audit Trail, System Implementation and Testing: Operational and Test Environment, Conversion Preparation, Database installation, users Training and Final Report to Management, Creating a new System, Test Plan: Activity Network for System Testing, Types of Testing.

Unit – V System Quality Assurance, IT infrastructure Selection and Evaluation of Processing and Maintenance Quality Assurance: Quality Factors specifications, Levels of Quality Assurance, Computer Hardware and Software Selection, Computer Configuration Determination, Requesting Proposal from Vendors, Evaluation of Vendor's Proposals, Acceptance of system. Evaluate of processing, Need of Maintenance.

Design and Analysis of Algorithm (C -505)

Unit – I Basic Concept of Algorithm: Definition of Algorithm, Characteristic of Algorithm, Pseudo Codes & Time Complexity of Basic Control Structures, Time and Space Complexity of insertion sort, Selection sort, Heap Sort, Bubble Sort, Asymptotic Notations (Growth of Functions)

Unit – II Divide and Conquer: Binary Search, Maximum & Minimum, Merge Sort, Quick Sort, Greedy Method General Method, Knapsack problem, Job sequencing with deadline, optimal Storage on tapes, Huffman Codes.

Unit – III Dynamic Programming: Matrix, Chain multiplications, Longest Common Subsequence, backtracking: General method, N Queens problem, Sum of Subsets.

Unit – IV Basic Traversals and Search Techniques, Techniques of binary trees, techniques of graph: BFS, DFS.

Unit – V Analysis of Graph Algorithm: Elementary Graph Algorithms, Multistage Graphs, Minimum Spanning Trees: Kruskal's & Prim's Algorithm, Single source shortest path, Dijkstra's & Bellman Ford, All Pairs shortest Path: Warshal Algorithm.

Note: In 6th Semester Students Takes Practical Training Only.

